MTH101, Basic Linear Algebra

Monsoon 2025

20/11/2025, 2:00PM-3:00PM.

Tutorial-9

Definition: For an $n \times n$ matrix A, the polynomial $\det(A - \lambda I_n)$ is called the *characteristic polynomial* of A and its roots are called the *eigenvalues* of A.

(1) Let A be an $n \times n$ matrix with real number entries and $T : \mathbb{R}^n \to \mathbb{R}^n$ be linear transformation defined by $Tx := Ax$ for all $x \in \mathbb{R}^n$. Calculate the dimension of $\ker T$ and $\operatorname{im} T$ for the following choices of A.

(i) $A = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{pmatrix}$

(ii) $A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

(2) Find the eigenvalues and eigenvectors of the following matrices

(i) $\begin{pmatrix} -7 & -2 & 10 \\ -3 & 2 & 3 \\ -6 & -2 & 9 \end{pmatrix}$

(ii) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{pmatrix}$

(3) For any $n \times n$ matrix A, show that A and A^T have the same characteristic polynomial (in particular, they have the same eigenvalues).

(4) If $v \in \mathbb{R}^n$ is an eigenvector of an $n \times n$ matrix A then v is also an eigenvector of A^m for every positive integer m.

(5) If A is an $n \times n$ matrix such that $\det A = 0$ then prove that 0 is an eigenvalue of A and furthermore, if $A^m = 0$ for some positive integer m then prove that 0 is the only eigenvalue of A.